Worksheet 6–2 A Riemann Sums #1
AP Calculus Name_____

Directions: Complete the following problems. Be sure to show all work.

x	1	3	4	8
$f(x)$	6	2	5	7

1. Use a right Riemann sum with the three subintervals indicated in the table above to approximate $\int_1^8 f(x)dx$.

2. Use a left Riemann sum with the three subintervals indicated in the table above to approximate $\int_1^8 f(x)dx$.

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x	0	1	2	5	8	10	12
$g'(x)$	2.1	4.7	1.9	5.6	8.2	4.3	2.5

3. Use a midpoint Riemann sum with the three subintervals indicated in the table above to approximate $\int_0^{12} g'(x)dx$.

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t	0	2	3	6
$L(t)$	3.5	2.0	2.1	1.5

4. After a basketball game, the rate that fans exit the school is modeled by the function L, where L is measured in fans per second and t is measured in seconds. Values of L are given in the table above for selected values of t Use a left Riemann sum with the three subintervals indicated in the table above to approximate the total number of fans that exited the school from $t = 0$ to $t = 6$ seconds. Include units of measure.

Time (hours)	0	2	3	6	10
$R(t)$ People/hour	37	44	36	42	48

5. The rate at which customers are being served at StarBruists is given by the continuous function $R(t)$. A table of selected values of $R(t)$, for the time interval $0 < t < 10$ hours, is given above. At $t = 0$, there had already been 200 customers served.

Use a trapezoidal sum with four subintervals to approximate how many customers had been served after 10 hours.

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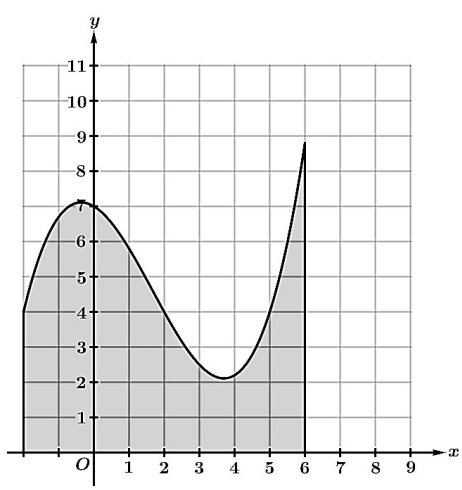
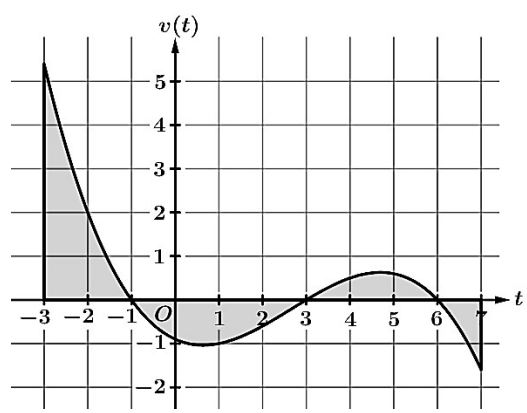
6. Approximate the area under the curve of $f(x) = -0.2x^2 - x + 12$ using a trapezoidal approximation on the interval $[-1, 3]$ with $n = 4$ equal subintervals.

The graph of a function is shown in each problem below. Write a definite integral expression that represents the shaded region.

7. graph of $h'(x)$



8. graph of $v(t)$